

AP STATISTICS MASTER REVIEW SERIES — BOOK 1

AP Statistics Formula & Inference Decision Guide

The Complete High-Yield Exam Companion: Formula Breakdowns, Mindmaps, Inference Decision Trees, and Rubric Sentence Frames

PR

2026–2027 Edition

Published for Independent Amazon KDP Distribution

© 2026–2027 PR. All rights reserved.

No part of this publication may be reproduced, distributed, or transmitted in any form or by any means, including photocopying, recording, or other electronic or mechanical methods, without prior written permission.

® Trademark Notice & Nominative Fair Use Disclaimer:

AP® and Advanced Placement® are trademarks registered by the College Board, which was not involved in the production of, and does not endorse, this product. TI-84 Plus® is a registered trademark of Texas Instruments. All trademarks are used under Nominative Fair Use for educational and descriptive purposes only.

Free Interactive Companion Web Portal:

Access searchable formula lookups, interactive inference wizards, and diagnostic drills:



Scan to Access Free Interactive Student Companion Portal

<https://eduprosuite-org.github.io/ap-stats-score5-blueprint/>

Published Independently via Amazon KDP

First Edition: 2026–2027

How to Use This Master Companion

Welcome to the **AP Statistics Formula & Inference Decision Guide**! This companion is engineered to replace bulky 600-page textbooks with high-yield clarity, visual mindmaps, and rubric sentence frames.

The 4-Pillar Master System

Pillar 1: Unit Mindmaps & Flowcharts: Master the visual architecture of each unit before diving into detailed formulas.

Pillar 2: Formula & Parameter Translations: Understand exactly what every Greek symbol ($\mu, \sigma, \hat{p}, \chi^2$) represents in plain English.

Pillar 3: Mnemonics & Grader Pitfalls: Use universal mnemonics (SOCS, BINS, LINER, PANIC, PHANTOM) to avoid losing partial credit.

Pillar 4: Fill-in-the-Blank Sentence Frames: Memorize standardized rubric templates to capture 100% credit on Free-Response Questions (FRQs).

Official AP[®] Statistics Exam Blueprint

1. Exam Format & Pacing Strategy (3 Hours / 180 Mins)

Section	Questions	Time	Weight	Pacing Strategy
Section I: MCQs	40 Questions	90 mins	50%	2.25 mins / question
Section II: FRQs	6 Questions	90 mins	50%	Q1–5: 13 mins each
<i>Part A (Q1–5)</i>	5 Standard	65 mins	37.5%	Core concepts
<i>Part B (Q6)</i>	1 Invest. Task	25 mins	12.5%	Novel applications

2. Official 9-Unit Syllabus Weightage Matrix

Unit	Curriculum Topic		Exam Weight	Yield Priority
Unit 1	Exploring Data	One-Variable	15–23%	Tier 1 (High)
Unit 2	Exploring Data	Two-Variable	5–7%	Tier 2 (Medium)
Unit 3	Collecting Data		12–15%	Tier 1 (High)
Unit 4	Probability, Random Variables & Dist.		10–20%	Tier 1 (High)
Unit 5	Sampling Distributions		7–12%	Tier 2 (Medium)
Unit 6	Inference for Proportions (Categorical)		12–15%	Tier 1 (High)
Unit 7	Inference for Means (Quantitative)		10–18%	Tier 1 (High)
Unit 8	Chi-Square Tests		2–5%	Tier 3 (Quick)
Unit 9	Inference for Slopes		2–5%	Tier 3 (Quick)

Exam-Day Morning Routine & Master Mnemonics

Exam-Day Morning Checklist

- **TI-84 Plus CE Inspection:** Charge your calculator to 100% the night before. Ensure RAM is clean and degree/radian mode does not interfere with list stats.
- **Digital Bluebook Readiness:** Verify your device is fully charged, software is updated, and your College Board login credentials are confirmed.
- **The 5-Minute Brain Dump:** When the exam starts, write down key mnemonics (SOCS, BINS, LINER, PANIC, PHANTOM) on your scratch paper.

The Top 5 Universal AP Statistics Mnemonics

1. SOCS — Describing Quantitative Distributions (Unit 1)

S — Shape: Skewed Left, Skewed Right, Symmetric, Unimodal, Bimodal.

O — Outliers: Apply $1.5 \times \text{IQR}$ rule or 2SD rule.

C — Center: Report Median (for skewed) or Mean (for symmetric).

S — Spread: Report IQR/Range (for skewed) or Standard Deviation (for symmetric).

Always include context and units!

2. BINS — Binomial Distribution Checks (Unit 4)

B — Binary: Only two possible outcomes (Success vs Failure).

I — Independent: Trials must be independent (or 10% condition holds).

N — Number: Fixed number of trials (n).

S — Same Probability: Probability of success (p) remains constant.

3. LINER — Linear Regression Inference Conditions (Unit 9)

L — Linear: Scatterplot shows linear pattern; residual plot shows random scatter.

I — Independent: 10% condition ($n \leq 0.10N$).

N — Normal Residuals: Histogram/normal probability plot of residuals is roughly normal.

E — Equal Variance: Residuals show equal spread across all $\hat{y}$ values.

R — Random: Data collected via random sampling or randomized experiment.

4. PANIC — 5-Step Confidence Interval Protocol

P — Parameter of interest defined in context (μ or p).

A — Assumptions & Conditions checked (Random, 10%, Normal / Large Counts).

N — Name the specific interval procedure (e.g., 1-Sample t -Interval for μ).

I — Interval calculated with formula and calculator degrees of freedom.

C — Conclusion stated using standard rubric sentence frames in context.

5. PHANTOM — 7-Step Hypothesis Testing Protocol

P — Parameter defined in context.

H — Hypotheses stated (H_0 and H_a using parameter symbols).

A — Assumptions checked.

N — Name the test procedure (e.g., 2-Prop z -Test).

T — Test statistic calculated (z , t , or χ^2).

O — Obtain p -value and report degrees of freedom.

M — Make decision: Reject H_0 or Fail to Reject H_0 with contextual conclusion.

Contents

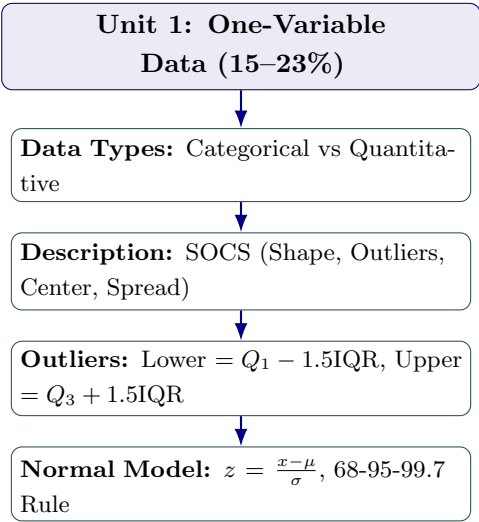
How to Use This Master Companion	ii
Official AP Exam Blueprint	iii
Exam-Day Routine & Master Mnemonics	iv
1 Unit 1: Exploring One-Variable Data	1
1.1 Unit 1 Visual Mindmap	1
1.2 Essential Formulas & Plain-English Translation	1
1.3 Resistant vs. Non-Resistant Summary Statistics	2
1.4 Top Exam Traps & Sentence Frames	2
2 Unit 2: Exploring Two-Variable Data	3
2.1 Unit 2 Visual Mindmap	3
2.2 Essential Formulas & Parameter Breakdowns	3
2.3 Sentence Templates for FRQs	4
3 Unit 3: Collecting Data	5
3.1 Unit 3 Visual Flowchart: Sampling vs. Experimentation	5
3.2 The 4 Principles of Experimental Design	5
4 Unit 4: Probability, Random Variables, & Distributions	7
4.1 Essential Formulas & Probability Rules	7
5 Unit 5: Sampling Distributions	8
5.1 Unit 5 Core Formulas & Central Limit Theorem	8

6 Unit 6: Inference for Proportions (Categorical Data)	9
6.1 1-Proportion vs. 2-Proportion Inference Procedures	9
7 Unit 7: Inference for Means (Quantitative Data)	10
7.1 1-Sample t , Paired t , and 2-Sample t Inference	10
8 Unit 8: Chi-Square Tests	11
8.1 The 3 Types of Chi-Square Tests	11
9 Unit 9: Inference for Slopes	12
9.1 Linear Regression Slope t -Test & t -Interval	12
10 Master Inference Selection Flowchart	13
11 Top 25 "Never-Forget" AP Grader Deduction Traps	15
12 Standard FRQ Interpretation Sentence Frames	17

Chapter 1

Unit 1: Exploring One-Variable Data

1.1 Unit 1 Visual Mindmap



1.2 Essential Formulas & Plain-English Translation

Unit 1 Core Formulas

1. Sample Mean ($\bar{x}$): $\bar{x} = \frac{\sum x_i}{n}$ — The arithmetic average; non-resistant to skew/outliers.

2. Sample Standard Deviation (s_x): $s_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$ — The typical distance of observations from the mean.

3. Standardized Score (z-score): $z = \frac{x - \mu}{\sigma}$ — Number of standard

deviations an observation falls above (+) or below (−) the mean.

1.3 Resistant vs. Non-Resistant Summary Statistics

Measure	Resistant to Outliers / Skew? Written to Use on Exam
Median & IQR	Yes Skewed data or data with outliers (Middle 50% only)
Mean & Standard Deviation	No Symmetric, bell-shaped distributions (Pulled toward the tail/extremes)

1.4 Top Exam Traps & Sentence Frames

Unit 1 Trap Alert: Comparative Language

When comparing two distributions on an FRQ, you **must** use comparative words ("greater than", "less than"). Stating two separate descriptions scores zero credit!

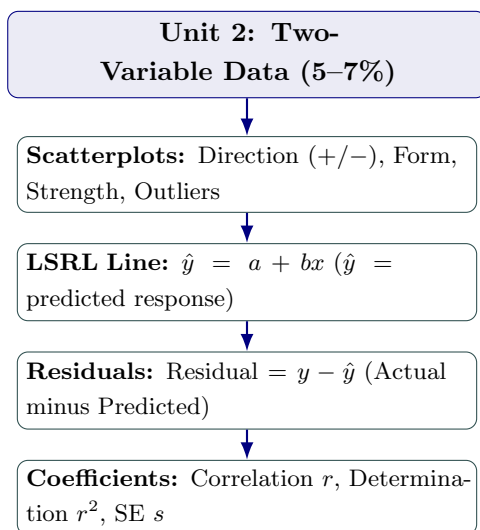
Unit 1 FRQ Sentence Frame: z-Score Interpretation

"The observed value of [context] is [z-score value] standard deviations [above/below] the mean of [context with units]."

Chapter 2

Unit 2: Exploring Two-Variable Data

2.1 Unit 2 Visual Mindmap



2.2 Essential Formulas & Parameter Breakdowns

Unit 2 Core Formulas: Linear Regression

- Least-Squares Regression Line:** $\hat{y} = a + bx$ ($\hat{y}$ = predicted response)
- Slope (b):** $b = r \left(\frac{s_y}{s_x} \right)$ — Change in predicted y per 1-unit increase in x .
- y -Intercept (a):** $a = \bar{y} - b\bar{x}$ — The point $(\bar{x}, \bar{y})$ is always on the LSRL.
- Residual (e):** Residual = $y - \hat{y}$ = Actual y – Predicted $\hat{y}$

2.3 Sentence Templates for FRQs

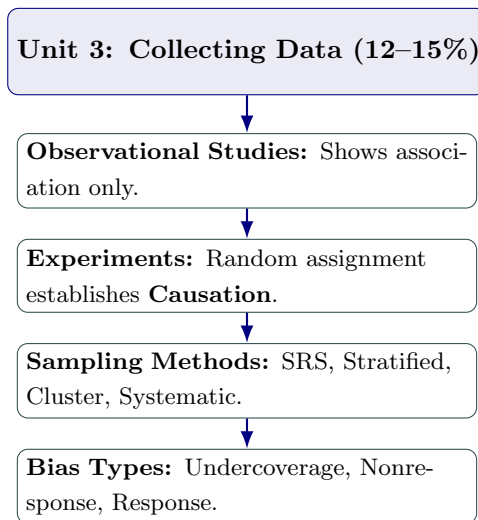
Unit 2 FRQ Sentence Templates

- 1. Slope (b):** *"For each additional 1 [unit of x], the predicted [response variable y] increases/decreases by approximately [value of b] [units of y]."*
- 2. r^2 :** *"Approximately [value of $r^2 \times 100\%$] of the variation in [response variable in context] is accounted for by the linear model with [explanatory variable in context]."*
- 3. Standard Error of Residuals (s):** *"The typical distance between the actual [response variable y] and the predicted values from the regression line is approximately [s units of y]."*

Chapter 3

Unit 3: Collecting Data

3.1 Unit 3 Visual Flowchart: Sampling vs. Experimentation



3.2 The 4 Principles of Experimental Design

1. **Comparison:** At least two treatment groups (or treatment vs. control) to isolate effects.
2. **Random Assignment:** Balances the effects of confounding variables across treatment groups.
3. **Control:** Keep all other outside variables constant.
4. **Replication:** Apply treatments to sufficiently many experimental units.

Unit 3 Trap Alert: Blocking vs. Stratification

Stratification is used when *sampling* from a population.

Blocking is used when *assigning treatments* in an experiment.

Never mix these terms on an AP exam!

Chapter 4

Unit 4: Probability, Random Variables, & Distributions

4.1 Essential Formulas & Probability Rules

Unit 4 Core Formulas

- 1. Addition Rule:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ (If disjoint, $P(A \cap B) = 0$).
- 2. Conditional Probability:** $P(A | B) = \frac{P(A \cap B)}{P(B)}$
- 3. Independence Test:** Events A and B are independent if $P(A | B) = P(A)$ or $P(A \cap B) = P(A) \cdot P(B)$.
- 4. Combining Random Variables:**

$$\mu_{X \pm Y} = \mu_X \pm \mu_Y$$

$$\sigma_{X \pm Y}^2 = \sigma_X^2 + \sigma_Y^2 \quad (\text{Always ADD variances!})$$

- 5. Binomial Distribution $B(n, p)$:**

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$\mu_X = np, \quad \sigma_X = \sqrt{np(1 - p)}$$

Unit 4 Trap Alert: Independence vs Disjoint

Mutually exclusive (disjoint) events can **NEVER** be independent (if $P > 0$).
Knowing A happened means B cannot happen!

Chapter 5

Unit 5: Sampling Distributions

5.1 Unit 5 Core Formulas & Central Limit Theorem

Sampling Distributions

1. Sample Proportion ($\hat{p}$):

$$\mu_{\hat{p}} = p, \quad \sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$$

Normal if $np \geq 10$ and $n(1-p) \geq 10$.

2. Sample Mean ($\bar{x}$):

$$\mu_{\bar{x}} = \mu, \quad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

Normal if $n \geq 30$ by CLT or population is normal.

Central Limit Theorem (CLT) Golden Rule

The CLT states that the **sampling distribution of $\bar{x}$** becomes approximately normal as n increases ($n \geq 30$), regardless of the shape of the underlying population! It does *not* apply to individual observations.

Chapter 6

Unit 6: Inference for Proportions (Categorical Data)

6.1 1-Proportion vs. 2-Proportion Inference Procedures

Unit 6 Formulas

1-Prop z -Interval: $\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

1-Prop z -Test: $z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$ (Use p_0 for Large Counts condition!)

2-Prop z -Test (Pooled):

$$\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2}$$

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}_c(1 - \hat{p}_c) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

Unit 6 PANIC Template for 1-Prop z -Interval

P: Let p = true proportion of [context].

A: Random sample; 10% condition ($n \leq 0.10N$); Large counts ($n\hat{p} \geq 10, n(1 - \hat{p}) \geq 10$).

N: 1-Proportion z -Interval.

I: Record [Lower, Upper] with 95% confidence.

C: "We are 95% confident that the interval from [Lower] to [Upper] captures the true proportion of [context]."

Chapter 7

Unit 7: Inference for Means (Quantitative Data)

7.1 1-Sample t , Paired t , and 2-Sample t Inference

Unit 7 Means Formulas

1-Sample t -Interval: $\bar{x} \pm t^* \left(\frac{s}{\sqrt{n}} \right)$ (df = $n - 1$)

1-Sample t -Test: $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$ (df = $n - 1$)

2-Sample t -Test: $t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$ (Pooled: NO)

Unit 7 Trap Alert: Paired t -Test vs. 2-Sample t -Test

If data comes from pre/post tests, twins, or the same experimental unit measured twice, it is a **Paired t -Test** on the single list of differences ($L_1 - L_2$). Never run a 2-sample t -test on paired data!

Chapter 8

Unit 8: Chi-Square Tests

8.1 The 3 Types of Chi-Square Tests

Test Name	Number of Samples	Variables	Degrees of Freedom
χ^2 Goodness of Fit	1 Sample	1 Categorical Variable	$df = k - 1$
χ^2 Homogeneity	2+ Independent Samples	1 Categorical Variable	$df = (r - 1)(c - 1)$
χ^2 Independence	1 Single Sample	2 Categorical Variables	$df = (r - 1)(c - 1)$

Chi-Square Test Statistic

$$\chi^2 = \sum \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}}$$

$$\text{Expected Count} = \frac{(\text{Row Total}) \times (\text{Column Total})}{\text{Table Total}}$$

Unit 8 Condition Alert

All **Expected Counts** must be ≥ 5 . Never check observed counts for this condition!

Chapter 9

Unit 9: Inference for Slopes

9.1 Linear Regression Slope t -Test & t -Interval

Unit 9 Formulas

t -Interval for Slope β : $b \pm t^* \cdot SE_b$ (df = $n - 2$)

t -Test for Slope β : $t = \frac{b-0}{SE_b}$ (df = $n - 2$)

LINER Condition Checklist for Slopes

- L** — Linear pattern in scatterplot.
- I** — Independent observations (10% rule).
- N** — Normal distribution of residuals.
- E** — Equal variance of residuals across line.
- R** — Random sampling or assignment.

Chapter 10

Master Inference Selection Flowchart

Data Type	Groups	Goal	Procedure Path
Categorical	1 Sample	Estimate p	1- STAT → TESTS → A Prop z- Interval
Categorical	1 Sample	Test p	1- STAT → TESTS → 5 Prop z- Test
Categorical	2 Samples	Estimate $p_1 - p_2$	2- STAT → TESTS → B Prop z- Interval
Categorical	2 Samples	Test $p_1 = p_2$	2- STAT → TESTS → 6 Prop z- Test (Pooled)
Categorical	1 Sample (Fit)	Fit dist.	χ^2 STAT → TESTS → D GOF- Test
Categorical	2+ Samples	Homogeneity	χ^2 STAT → TESTS → C Test of Ho- mo- gene- ity
Categorical	1 Sample (2 Var)	Association	χ^2 STAT → TESTS → C Test of In- de- pen- dence
Quantitative	1 Sample	Estimate μ	1- STAT → TESTS → 8 Sample t- Interval
Quantitative	1 Sample	Test μ	1- STAT → TESTS → 2 Sample t- Test
Quantitative	Matched Pairs	Difference μ_d	Paired Test on $L_1 - L_2$ t- Test/Interval
Quantitative	2 Indep Samples	Compare $\mu_1 - \mu_2$	2- STAT → TESTS → 4/0 Sample t-

Chapter 11

Top 25 "Never-Forget" AP Grader Deduction Traps

1. **NEVER** say "Accept H_0 ": Always write "*Fail to reject H_0* ".
2. **NEVER** write "We prove H_a ": Write "*We have convincing statistical evidence for H_a* ".
3. Hypotheses in parameters (μ, p, β) , **NEVER** statistics $(\bar{x}, \hat{p}, b)$.
4. Always define parameters in context with units.
5. Comparative language required on all distribution comparison FRQs (greater, lesser).
6. Always put a hat ($\hat{y}$) on regression predicted values.
7. Correlation (r) measures **LINEAR** strength only (never use r for curved data).
8. Association is **NOT** Causation.
9. Blocking is for experiments; Stratification is for sampling.
10. Mutually exclusive events are **NEVER** independent.
11. Always **ADD** variances (σ^2), **NEVER** add standard deviations (σ).
12. CLT applies to the sampling distribution of $\bar{x}$, not sample data.
13. Use hypothesized p_0 for 1-Prop z -Test Large Counts ($np_0 \geq 10, n(1 - p_0) \geq 10$).
14. Use sample $\hat{p}$ for 1-Prop z -Interval Large Counts ($n\hat{p} \geq 10, n(1 - \hat{p}) \geq 10$).
15. Chi-square conditions require **EXPECTED** counts ≥ 5 , not observed counts.

16. Chi-square df: $df = (\text{rows} - 1)(\text{cols} - 1)$.
17. Never pool standard deviations for 2-sample t -tests (Pooled: NO).
18. Confidence Interval (plausible values) vs Confidence Level (long-run capture rate).
19. A high r^2 does NOT prove linearity (always check residual plot for random scatter).
20. Voluntary response produces severe bias; results cannot be generalized.
21. State test conclusion in terms of H_a in context.
22. State the significance level (α) explicitly in decision sentences.
23. Calculator Speak alone scores ZERO on FRQs (must write formulas and parameter names).
24. Extrapolation is dangerous (do not predict $\hat{y}$ outside the domain of x).
25. Context, Context, Context: Include problem units and scenario in every sentence!

Chapter 12

Standard FRQ Interpretation Sentence Frames

Frame 1: Confidence Interval for a Parameter

"We are [C%] confident that the interval from [Lower] to [Upper] captures the true [proportion / mean] of [context with units]."

Frame 2: Confidence Level (C%)

"If we take many random samples of the same size and construct [C%] confidence intervals, about [C%] of the intervals will capture the true [parameter in context]."

Frame 3: p -Value in Context

"Assuming that [null hypothesis H_0 in context] is true, there is a [p-value] probability of obtaining a sample statistic as extreme as or more extreme than observed, purely by random chance."

Frame 4: Reject H_0 Conclusion

"Because the p -value ([p]) is less than α ([0.05]), we reject H_0 . There is convincing statistical evidence that [state H_a in context]."

Frame 5: Fail to Reject H_0 Conclusion

"Because the p -value ([p]) is greater than α ([0.05]), we fail to reject H_0 . There is NOT convincing statistical evidence that [state H_a in context]."